

Title: SOP – Transfer UPS from Maintenance Bypass Mode to Normal (Online) Mode**Standard Operating Procedure****Version 2.1****Modified _____**

Label	Content
Title	Transfer UPS from Maintenance Bypass Mode to Normal (Online) Mode
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Data Center Operations**Procedure TITLE:** Transfer UPS from Maintenance Bypass Mode to Normal (Online) Mode**Procedure #:** NA_US_ Transfer UPS from Maintenance Bypass Mode to Normal (Online) Mode**Site(s) Designation:****SOP (STANDARD OPERATING PROCEDURE) DOCUMENT**

- This Standard Operating Procedure (SOP) details a common operating procedure and can be referenced whenever needed. The document should be used for mode transfer of a single UPS System.

DESCRIPTION OF WORK

- This SOP will be used to transfer UPS system to Normal (Online) Mode for activities including maintenance, repairs, upgrades, and testing.

- The technician will interact with the following switchboards and panels to transfer UPS modes and verify conditions and values:
 - Uninterruptible Power Supply (UPS) System
 - Maintenance Bypass Switchboard (MBS)
 - Advanced Automatic Switchboard (AASB) – not operated during this procedure
 - Battery Cabinet (BC)
 - Power Distribution Unit (PDU)

PRECAUTIONS

- While the UPS is in Maintenance Bypass Mode:
 - The critical load will be entirely supported by utility (standard) or generator power, and the battery backup will be unavailable. Any interruption in utility power will result in a loss of one PDU power source to the critical load.
 - Avoid transferring from utility power to generator power or vice versa, if possible. This action will result in a brief loss of one source of power to the critical load.

Estimated Time to Complete

- ____1.0____ Hours (Add in half hour increments)

List of Affected Equipment

Fill in the blanks with the specific details for the UPS System being isolated.

Equipment Type:	Equipment Label or Asset ID:	Location:
UPS	1-UPS-_____	Electrical Room _____
Switchgear	1-MBS-_____	Electrical Room _____
PDU	1-PDU-_____	Electrical Room _____

Safety Requirements

- Perform a Job Hazard Analysis (JHA) and attach the document to the Work Order.

Time	Procedure	Check Box	Completed Initials
	1. INFORM the following personnel that UPS will be placed into Normal (Online) Mode and that these alarms are expected on BMS during the performance of the procedure: a. UPS on Bypass b. MBS on Bypass		
	1. Energy	☐	
	2. Facilities Management	☐	
	3. OCC	☐	
	2. CONDUCT a pre-evolution brief that includes at a minimum the operators performing the procedure, the OCC operator, and all vendors involved in the maintenance. ENSURE the brief includes a review of the procedure, each person's roles and responsibilities, and a review of the associated JHA.	☐	
	3. VERIFY there are no abnormal alarms on the following control panels:		
	a. UPS Control Panel HMI (Logs Tabs)	☐	
	b. PDU HMI	☐	
	c. AASB HMI #1 and #2	☐	
	4. VERIFY there are no abnormal alarms on the generators for the lineup associated with the UPS being placed into Normal (Online) Mode.	☐	

	Electrical Room _____		
	5. ATTACH a magnetic red LED light to the MBS, UPS and Battery Cabinet being operated and ENERGIZE the lights.	☐	
	6. ESTABLISH or VERIFY the electrical safety work boundaries.	☐	
	7. Prior to transferring the UPS to Normal (online) mode. PERFORM a closeout inspection as follows:		
	a. VERIFY all tools and debris are removed from inside and around UPS.	☐	
	b. VERIFY UPS covers and hardware are properly installed.	☐	
	8. TURNOVER the UPS system from the technician to Operations. The vendor or technician signature indicates that all repair or maintenance work is completed, and the UPS is ready for restoration to online service: a. Technician Name: _____	☐	
	b. Technician's Company: _____		
	c. Technician Signature: _____		
	Prepare the UPS for Power Restoration		

	9. REMOVE LO/TO from the following affected units	☐	
	a. PDU Sub feed 17	☐	
	b. Battery cabinet strings 1-6 (unlock and open cabinets)	☐	
	10. REMOVE the LOTO from the Battery String circuit breakers.	☐	
	a. CLOSE circuit breaker Sub feed 17 (Battery String Control Power).		
	b. CLOSE each of the Battery String circuit breakers.		
	1. Battery String Circuit Breaker 1.	☐	
	2. Battery String Circuit Breaker 2.	☐	
	3. Battery String Circuit Breaker 3.	☐	
	4. Battery String Circuit Breaker 4.	☐	
	5. Battery String Circuit Breaker 5.	☐	
	6. Battery String Circuit Breaker 6.	☐	
	7. Battery String Circuit Breaker 7 (if present).	☐	
	c. CLOSE and LOCK the battery string cover panels.	☐	
	11. REMOVE the LOTO from the MIS and MIB circuit breakers.		

	a. MIS circuit breaker.	☐	
	b. MIB circuit breaker. If the MIB circuit breaker LOTO was already removed, then mark “N/A.”	☐	
	Place the UPS in Static Bypass (if not already done)		
	12. On the MBS, CLOSE the MIB circuit breaker. If the MIB circuit breaker was already closed for UPS testing, then mark “N/A.”	☐	
	Note: Allow time for UPS to start up and monitor the unit for abnormalities as it comes online. The UPS will automatically start up into Static Bypass mode.		
	13. On the UPS Control Panel, RECORD the following: a. UPS Mode: _____ (Static Bypass or Startup Mode) b. UPS on Inverter(s): _____ (Yes or No) c. UPS Load kVA: _____ kVA d. No unexpected Active Events on the Event Tab	☐	
	14. If the UPS is already in Static Bypass mode, then go to Step 17.e.		
	15. On the UPS Control Panel, perform the following to place the UPS in Static Bypass:		
	a. PRESS the padlock graphic in the upper right-hand corner of the screen and ENTER the level 2 password 0101 to unlock the screen.	☐	
	b. PRESS the “Controls” button to access the System Controls screen.	☐	

	c. On the System Controls screen, SELECT “Go To Bypass.”	☐	
	d. When the Bypass Control Command screen pops up, PRESS “Yes” to confirm.	☐	
	e. VERIFY the following on the UPS Control Panel HMI:		
	1. The “Bypass” status indicator is illuminated (amber).	☐	
	2. The “Bypass” status indicates On.	☐	
	1. The UPS Output kVA recorded previously is now indicated on the Bypass Meters:		
	a. PRESS the “Meters” button.	☐	
	b. PRESS the “Bypass Meters” button.	☐	
	c. PRESS the “Show Details” button.	☐	
	d. UPS Bypass – Phase A: _____ kVA UPS Bypass – Phase B: _____ kVA UPS Bypass – Phase C: _____ kVA UPS Bypass – Total: _____ kVA	☐	
	1. PRESS the “Home” button.	☐	
	i. VERIFY the following on the UPS lineup graphic:	☐	

	1. Rectifiers are deenergized (no green box). 2. Batteries are not charging (no green down arrow). 3. Inverter output contactors are open.		
	e. PRESS the “Meters” button.	☐	
	Transfer the UPS to Normal (Online) Mode		
	16. At the MBS, perform the following to place the UPS in Normal (Online) Mode:		
	a. VERIFY the SKRU Lamp is illuminated (white) to indicate the UPS is in Static Bypass.	☐	
	b. On the SKRU, PRESS and HOLD the white push button. TURN and REMOVE Key A, then RELEASE the pushbutton.	☐	
	c. INSERT Key A into the MBP circuit breaker Kirk Key Interlock and TURN clockwise to retract the bolt to unlock the MIS circuit breaker.	☐	
	d. CLOSE the MIS circuit breaker.	☐	
	a. At the UPS Control Panel, VERIFY that minimal load has transferred to the MIS circuit breaker by checking the following: 1. Load on the Bypass Meters is 0 kVA. 2. Home screen graphic indicates Maintenance Bypass and Static Bypass lines are paralleled (both green).	☐	
	b. REMOVE Key B from the MIS Kirk Key Interlock.	☐	
	c. INSERT and TURN Key B into the MBP circuit breaker Kirk Key Interlock.	☐	

	<u>WARNING:</u> Failure to close the MIS breaker prior to opening the MBP breaker will result in a loss of one source of power to the critical load.		
	d. OPEN the MBP circuit breaker.	☐	
	e. TURN and REMOVE Key A to lock MBP circuit breaker in the Off position.	☐	
	f. On the SKRU, INSERT and TURN Key A to lock the MBS into Normal mode.	☐	
	17. On the UPS Control Panel, perform the following to complete the transition to the Normal (Online) Mode:		
	a. PRESS the padlock graphic in the upper right-hand corner of the screen and ENTER the level 2 password 0101 to unlock the screen.	☐	
	b. PRESS the “Controls” button to access the System Status screen and SELECT “Go To Online.”	☐	
	c. When the Confirmation screen pops up, PRESS “Yes” to confirm.	☐	
	d. VERIFY the following on the UPS Control Panel:		
	1. The “Normal” status indicator is illuminated (green).	☐	
	2. The UPS status indicates Online.	☐	
	3. The four (4) UPM statuses each indicate Online.	☐	
	18. On the UPS Control Panel, RECORD the following: a. UPS on Inverter: _____ (Yes or No) b. UPS Output kVA: _____ kVA c. UPS Output kW: _____ kW	☐	

	d. No Active Events on the Events Tab. e. Verify all abnormal alarms are cleared.		
	19. VERIFY there are no abnormal alarms on the following control panels:		
	a. UPS Control Panel	☐	
	b. PDU HMI	☐	
	c. AASB HMI #1 and #2	☐	
	d. On the MBS, the Bypass status indicator is unlit (not glowing white).	☐	
	20. DOFF arc flash PPE.	☐	
	21. DEENERGIZE and REMOVE the magnetic red LED lights from the MBS, UPS and Battery Cabinet and RETURN them to their normally stowed location.	☐	
	22. TAKE DOWN the electrical safety work boundaries.	☐	
	Communicate Status Change of UPS		
	23. INFORM the following personnel that UPS has been placed into Normal (Online) Mode.	☐	
	a. Client	☐	
	b. Facilities Management	☐	
	c. OCC	☐	
TIME	BACKOUT PROCEDURES	Check	Completed Initials

		Box	
	1. Backout procedures will commence immediately if any problem occurs that affects the critical load.	☐	
	2. In the event of equipment failure or critical power outage:	☐	
	a. Stop all work.	☐	
	b. Safely clear all tools and personnel from the work area.	☐	
	3. INITIATE the Escalation Procedure, as appropriate.	☐	
EVENT COMPLETION			
SOP SUPERVISOR	TITLE:	SIGNATURE	DATE
Comments/lesson Learned			